

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States mean = 40 hours	AO1.2	B1	Mean = $\frac{1}{\lambda} = 40$ hours
(b)	Obtains correct probability	AO1.1b	B1	$P(\text{time} < 12) = 1 - e^{-12 \times 0.025}$ $= 1 - e^{-0.3} = 0.259$
(c)	Uses 'no memory' property PI	AO3.4	M1	Exponential distribution has no memory
	Obtains probability	AO1.1b	A1	$P(\text{time} > 30) = e^{-30 \times 0.025} = e^{-0.75}$ $= 0.472$
(d)	States or uses new mean (or uses $e^{-0.3}$)	AO3.4	B1	4 consecutive shifts gives $4 \times 12 = 48$ hours
	Finds probability $P(\text{time} > 48)$ (or uses $(e^{-0.3})^4$)	AO1.1a	M1	$P(\text{time} > 48) = e^{-48 \times 0.025} = e^{-1.2}$
	Obtains correct probability	AO1.1b	A1	$= 0.301$
(e) (i)	States Poisson for model of situation given	AO3.3	B1	Poisson identified
	States value for λ (= 0.025 per hour) for model	AO3.3	B1	Po(0.025) per hour
(ii)	Uses $60 \times$ 'their' λ for 'their' Po model Or Uses exponential model to find $P(\text{Time} > 60)$	AO3.4	M1F	For 60 hours of process $\lambda = 60 \times 0.025 = 1.5$
	Obtains correct probability using Poisson or exponential model	AO1.1b	A1	$P(X=0) = \frac{1.5^0 \times e^{-1.5}}{0!} = e^{-1.5} =$ 0.223
				Total 11 marks

Q2.

(a) (i)

$$\mu = E(X) = \int_0^{\infty} kxe^{-kx} dx$$

M1

$$= [-xe^{-kx}]_0^{\infty} + \int_0^{\infty} e^{-kx} dx$$

M1

$$= 0 + \left[\frac{e^{-kx}}{-k} \right]_0^{\infty}$$

A1

$$= \frac{1}{k}$$

AG

3

$$(ii) \int_0^m ke^{-kx} dx = \frac{1}{2}$$

$$\text{Or } F(x) = 1 - e^{-kx}$$

M1

$$[-e^{-kx}]_0^m = \frac{1}{2}$$

$$-e^{-km} + 1 = \frac{1}{2}$$

$$\frac{1}{2} = 1 - e^{-km}$$

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A1

$$e^{-km} = \frac{1}{2} \Rightarrow km = \ln 2 \Rightarrow m = \frac{\ln 2}{k}$$

M1A1

$$\ln 2 < 1 \Rightarrow m < \mu$$

A1

5

$$(b) (i) P(T = 0) = e^{-\frac{t}{\lambda}}$$

B1

1

$$(ii) (A) P(T > t) = 1 - F(t) = e^{-\frac{t}{\lambda}}$$

M1

$$P(T < t) = F(t) = 1 - e^{-\frac{t}{\lambda}} \quad t \geq 0$$

AG

A1

2

(B) $f(t) = F'(t) = \frac{1}{\lambda} e^{-\frac{t}{\lambda}} \quad t \geq 0$

M1

$\Rightarrow$ Exponential distribution

A1

2

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Q3.

(a) $D \sim \text{Exp}(16)$

Standard deviation, $\sigma = 16$

cao

B1

1

(b) (i) Exponential with $\frac{1}{\lambda} = 16$

Use of associated pdf or df

M1

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$$P(D > 20) = \left[-e^{-\frac{t}{16}} \right]_{20}^{\infty}$$

or

$$= 1 - \left(1 - e^{-\frac{20}{16}} \right)$$

Either

A1

$$e^{-1.25} = 0.286 \text{ to } 0.287$$

awfw

A1

(ii) $P(20 < D < 30) = e^{-1.25} - e^{-1.875}$

awfw

A1

$$= 0.28650 - 0.15335 = 0.133$$

awrt

A1

5

- (c) $P(D_2 > 20 | D_1 > 20) =$
Use of conditional probability

M1

$$\frac{P(D_1 + D_2 > 20 | D_1 > 20)}{P(D_1 > 20)}$$

Correct expression

A1

$$e^{-25} \div e^{-125} = e^{-125} = 0.286 \text{ to } 0.287$$

ft if calculated answer = (b)(i)
Justification of equality $\Rightarrow$ 3 when (b)(i) correct
Justification of equality $\Rightarrow$ 2 when (b)(i) incorrect

A1.ft

3

- (d) Poisson

B1

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$$\text{Parameter/mean} = \frac{5}{16}$$

$$= 0.312 \text{ to } 0.13$$

cao/awfw

B1

2

[11]

Examiner reports

Q2.

The method marks in parts (a)(i) and (ii) were generally earned. Almost inevitably there were errors with signs, integration and use of logarithms, costing accuracy marks. Most candidates stated the correct probability in part (b)(i), but failed to recognise that it was needed in part (b)(ii)(A). The hope that 2 marks could be gained by writing down what was on the question paper was forlorn. Candidates needed to state that $P(T > t)$ was the probability that 0 strikes occur in the interval $(0, t)$ and hence that $P(T > t) = F(t) = 1 - e^{-t/\lambda}$ for $t \geq 0$. Most candidates knew that they needed to differentiate F with respect to t in order to obtain $f(t)$, but some then forgot to state the distribution of T , which was asked for in the question.

Q3.

It was disappointing to see the large proportion of candidates who could not score the first mark despite the relevant information being stated clearly on page 14 of the formulae booklet. Nevertheless, most candidates scored most, often all, of the 5 marks available in part (b). In part (c), very few candidates appeared aware of the 'non-memory' property of an exponential distribution. Of the few who scored marks, less than half quoted their answers to part (a)(i) with most starting a new using conditional probability with varying degrees of success. In part (d), about 50% of candidates scored 1 mark for either

"Poisson" or " $\frac{5}{16}$ ", but very few indeed scored the 2 marks available.